

Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 7A nr. 3.3

$$\begin{aligned} & \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 + y^4}{(x^2 + y^2)^{\frac{3}{2}}} \\ & x = r \cos \theta \text{ en } y = r \sin \theta \\ & = \lim_{r \rightarrow 0} \frac{(r \cos \theta)^4 + (r \sin \theta)^4}{(r^2 \cos^2 \theta + r^2 \sin^2 \theta)^{\frac{3}{2}}} \\ & = \lim_{r \rightarrow 0} \frac{r^4 (\cos^4 \theta + \sin^4 \theta)}{(r^2 (\cos^2 \theta + \sin^2 \theta))^{\frac{3}{2}}} \\ & = \lim_{r \rightarrow 0} \frac{r^4 (\cos^4 \theta + \sin^4 \theta)}{(r^2 \cdot 1)^{\frac{3}{2}}} \\ & = \lim_{r \rightarrow 0} \frac{r^4 (\cos^4 \theta + \sin^4 \theta)}{r^3} \\ & = \lim_{r \rightarrow 0} r (\cos^4 \theta + \sin^4 \theta) \\ & = 0 \end{aligned}$$

Want sin en cos zijn begrensde functies.

References